# Predicting Housing Affordability Stress Across Greater Boston ZIP Codes

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## Motivation

Housing affordability in Greater Boston has deteriorated sharply since 2020. Rising prices, constrained supply, and volatile mortgage rates have pushed homeownership out of reach for growing numbers of residents, but the pressure is not uniform across neighborhoods.

**Goal:** Predict which ZIP codes will experience affordability stress *before* it materializes, giving policymakers and prospective buyers an early warning signal.

**Definition:** A ZIP – month is *high stress* when the income required to buy a median home falls in the **top 25%** of all observations in the dataset.

## Data

- **Zillow ZHVI:** Monthly ZIP-level home values across the Boston metro area, seasonally adjusted
- **FRED:** 30-year fixed mortgage rate, national unemployment rate, and Consumer Price Index
- **Zillow market signals:** Metro-level inventory counts and median days to pending sale
- **Target:** Zillow income needed to purchase, thresholded at the 75th percentile for binary labels

## Three Modeling Approaches

### 1 SARIMAX

*Time Series Forecasting*

Order $(1,1,1)(1,1,1)_{12}$ selected via ACF/PACF across six representative ZIP codes. 30-year mortgage rate as exogenous regressor. Trained through 2022; evaluated on 2023 – 2024.

### 2 MLP Classifier

*Binary Stress Prediction*

Built from scratch in NumPy. 284 inputs → 32 (ReLU) → 16 (ReLU) → 1 (sigmoid). Learning rate 0.005, 600 epochs, binary cross-entropy loss. 3:1 class imbalance corrected via downsampling.

### 3 Bayesian Regression

*Interpretable Inference*

Implemented in R with `brms`/Stan. Predicts income needed to purchase a home with full posterior uncertainty over each predictor. Trained 2018 – 2023; evaluated 2024.

## References

[1] Truong et al. (2020). Housing price prediction via improved machine learning techniques. *Procedia Comp. Sci.*

[2] Zhang et al. (2023). Housing price prediction using ML: Wenzhou as a case study. *IJERPH.*

[3] Muralidharan & Minu (2015). Regression and neural network models for prediction of house prices. *J. Intell. Fuzzy Syst.*

[4] Zhang (2021). Random Forest for Boston house price prediction. *J. Physics: Conf. Series.*

[5] Gupta, Kabundi & Miller (2011). Forecasting the US real house price index. *Econ. Modelling.*

## 1 SARIMAX Forecasts

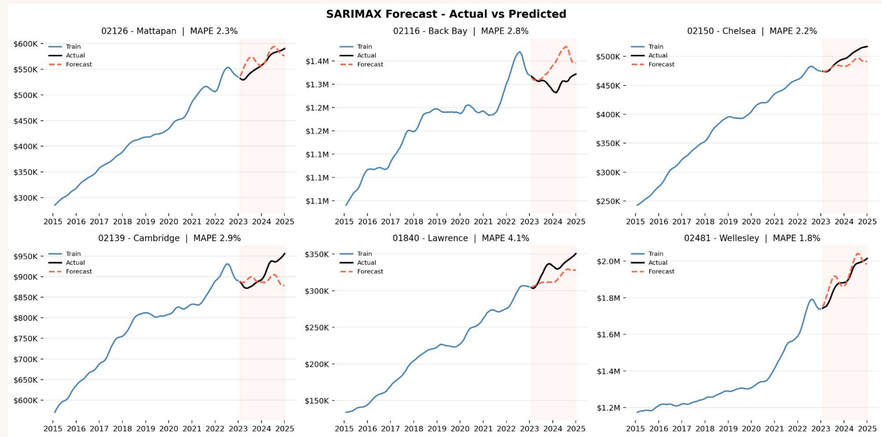


Fig. 1: Actual vs. forecast for six ZIP codes spanning Boston City, inner suburbs, and outer suburbs.

| ZIP Code | MAPE | MAE |
|---|---|---|
| 02481 Wellesley | 1.8% | $32,510 |
| 02150 Chelsea | 2.2% | $10,989 |
| 02126 Mattapan | 2.3% | $12,612 |

| ZIP Code | MAPE | MAE |
|---|---|---|
| 02116 Back Bay | 2.8% | $35,860 |
| 02139 Cambridge | 2.9% | $25,041 |
| 01840 Lawrence | 4.1% | $12,267 |

## 2 MLP Classifier: Stress Across Neighborhoods

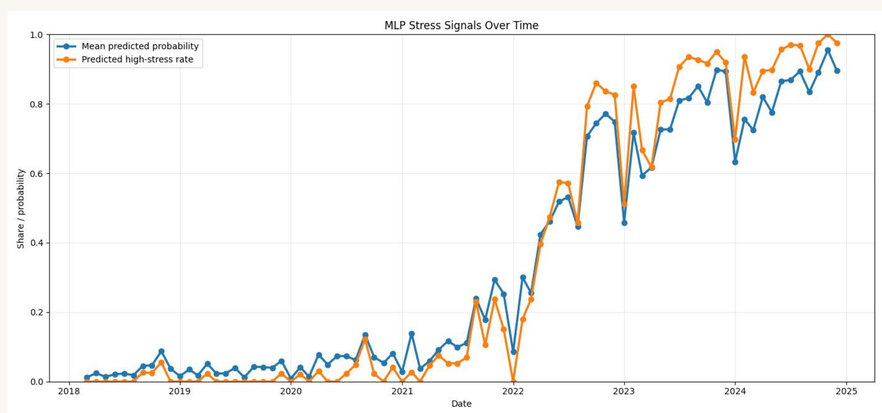


Fig. 2: Mean predicted stress probability (blue) and share of ZIP codes classified high stress (orange).

Both signals hover near zero through 2020, rise in 2021, and climb sharply through 2022 – 2023. By late 2024 the model flagged ∼**90% of ZIP codes** as high stress, including historically accessible neighborhoods like **Mattapan, Dorchester, and Roxbury Crossing** that lower and middle income residents have traditionally relied on.

**Confusion Matrix**

| | Pred. Low | Pred. High |
|---|---|---|
| **True Low** | 2,092 | 378 |
| **True High** | 82 | 786 |

**Classification Metrics**

| | |
|---|---|
| Accuracy | **86.2%** |
| Precision | **0.68** |
| Recall | **0.91** |
| F1 | **0.77** |

## Key Finding

**Each +1 pp in the mortgage rate adds ∼$10,000 to the income needed to buy a home.**

Borrowing costs, not home prices alone, are the primary driver of affordability stress in Greater Boston. All three models converge on this conclusion.

## 3 Bayesian Regression

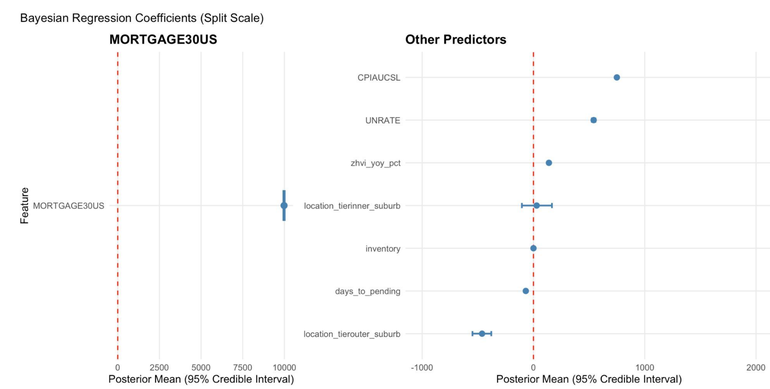


Fig. 3: Posterior means with 95% credible intervals. Mortgage rate dominates all others by an order of magnitude.

| | |
|---|---|
| Test MAE | **$4,306** |
| Test RMSE | **$4,481** |
| Test MAPE | **2.35%** |

## Conclusions & Future Work

The affordability crisis is no longer confined to expensive neighborhoods like Back Bay or Cambridge. It has spread into areas that lower and middle income residents have historically relied on. By late 2024 the MLP classifier flagged nearly **90% of ZIP codes** as high stress. The Bayesian model pinpoints the mechanism: each percentage point on the mortgage rate translates to roughly **$10,000 more income** required to buy a home, with high posterior certainty. The SARIMAX forecasts confirm the shift was large enough to challenge even well-specified time series models, particularly in high-value markets that experienced sharp post-2022 corrections.

A return to lower mortgage rates would be the single most impactful lever for easing stress, but without increased housing supply the underlying price pressure will persist.

**Limitations.** All three models predict *market-level* pressure but cannot identify which households within a ZIP code are at risk or have already been displaced. The income-needed variable is metro-level, flattening real variation across neighborhoods.

**Future work.** Incorporate ZIP-level demographic and income data to predict whether current residents can *afford to stay*, not just whether a neighborhood is expensive. Test Bayesian model robustness on a pre-2018 window to confirm that the mortgage rate dominance is not specific to the post-pandemic period.